

Multiplicity adjustment for Bayesian hypothesis testing via the probability of direction

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Abstract

The probability of direction (pd) has been rapidly adopted in psychological research as an intuitive Bayesian index for quantifying the direction of an effect. In practice, however, pd is routinely applied as a dichotomous decision rule using fixed thresholds (e.g., $pd > 0.975$), and when used across multiple parameters simultaneously, this practice inflates the probability of drawing misleading conclusions. We argue this problem is rooted in neglecting how probable it is, a priori, that any given effect is truly null. To address this, we propose a prior-odds adjustment to pd that explicitly encodes prior beliefs about the prevalence of null effects. This approach accounts for multiplicity without modifying the original model, offers a transparent way to encode prior skepticism about the existence of effects, and reduces the risk of false positives in multi-parameter settings. We evaluate its behavior across varying sample sizes and null proportions, and provide an R package for applied use.

Keywords: probability of direction, multiple testing, prior-odds

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Over the past two decades, the use of Bayesian methods in psychological research has grown considerably (Jevremov & Pajić, 2024; Pfadt et al., 2025; Van De Schoot et al., 2017). This expansion has, however, been accompanied by practices reminiscent of the “null ritual” in frequentist statistics, where widespread

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Code is openly available in the *multibayes* repository on GitHub at <https://github.com/mar-cald/multibayes>. The authors declare that they have no competing interests.

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adoption gave way to automatic application rather than principled reasoning (Cohen, 1994; Gigerenzer, 2004). For instance, researchers sometimes treat Bayes Factor thresholds (e.g., $BF > 3$) as dichotomous decision rules, or claim that an effect “exists” because the posterior probability of a directional effect exceeds a conventional benchmark such as 95% (Heck et al., 2023; Hoijtink et al., 2016; Kelter, 2023; Kruschke, 2018; Tendeiro et al., 2024; Tendeiro & Kiers, 2019). While such summaries are convenient, their use as default cutoffs encourages all-or-none interpretations. When these binary judgments are applied repeatedly across many parameters, they generate a multiplicity problem no different in kind from the one long recognized in frequentist statistics.

Yet even there, the problem of multiplicity has been handled inconsistently in practice. Researchers typically account for multiplicity in well-recognized cases, such as post hoc comparisons, but rarely do so when evaluating several effects simultaneously within the same model (Zeevi et al., 2025). This is routine in ANOVA or multiple regression, where main effects, interactions, and multiple predictors each constitute a distinct hypothesis test. Either way, the probability of at least one false positive accumulates with the number of tests performed, regardless of how those tests are framed or grouped (Bender & Lange, 2001; Cramer et al., 2016). Frequentist methods have been developed to address precisely this accumulation, targeting either the Family-Wise Error Rate (FWER; the probability of at least one false positive) or the False Discovery Rate (FDR; the expected proportion of false positives among rejected hypotheses) (Benjamini & Hochberg, 1995; Westfall & Young, 1993). These procedures address whether a spurious result enters the literature, but leave unresolved a subtler consequence of the same selection process. When researchers examine many parameters but highlight only the most extreme estimates, selection systematically inflates reported magnitudes (Altoè et al., 2020; Gelman & Carlin, 2014; Zwet & Cator, 2021).

Bayesian methods can mitigate this overestimation by pulling extreme estimates toward more plausible values, via informative priors (Dawid, 1994; Zwet & Gelman, 2022) and hierarchical modelling (Berry & Hochberg, 1999; Gelman et al., 2012, 2013; Kruschke, 2011). Crucially, however, magnitude bias and false positive accumulation are related but distinct problems. Shrinkage does not, on its own, provide any guarantee about the error properties of a repeated decision rule applied across many parameters: if the same cutoff is applied to many parameters (e.g., declaring an effect whenever the posterior probability exceeds 95%), the probability of spurious findings can remain substantial. Bayesian solutions to this issue typically rely on either simultaneous credible statements (Box & Tiao, 1992; Gelman & Tuerlinckx, 2000; Pruzek, 2016; Schnell et al., 2020) or priors over the hypothesis space that scale with the number of tests (Jeffreys, 1938; Scott & Berger, 2006; Scott & Berger, 2010; Westfall et al., 1997). Despite these available tools, hierarchical modeling alone is still routinely treated as sufficient for multiplicity control, a conflation that leaves the problem only partially addressed (Chen et al., 2020; Erčulj & Štrumbelj, 2015; Hogeveen et al., 2022; Limbachia et al., 2021; Mathar et al., 2022).

This issue becomes particularly salient when posterior summaries such as the probability of direction (pd) are used as decision rules. The pd is a posterior index that quantifies the probability that an effect lies in its dominant direction (Makowski et al., 2019). In recent years, it has increasingly been adopted as a standardized decision criterion, with researchers claiming evidence for an effect when it exceeds fixed thresholds (e.g., $pd > 0.95$ or $pd > 0.975$) (Ben-Artzi et al., 2022; Bramley & Ruggeri, 2022; Brissenden et al., 2023, 2024; Ciranka et al., 2026; Hochman et al., 2025; Makowski et al., 2020; Schurr et al., 2024; Stone-Sabali et al., 2024). A key property of pd is its formal correspondence with the frequentist p -value: for location parameters in exponential family models under a uniform prior, pd equals the complement of the one-sided p -value, and this relationship holds approximately under a broader class of priors (Casella & Berger, 1987; Marsman & Wagenmakers, 2017). Because weakly informative or flat priors are commonly used in applied psychological research (Smid & Winter, 2020; Van De Schoot et al., 2017), thresholding pd effectively reinstates a decision procedure that is susceptible to multiplicity inflation in much the same way as classical significance testing. Yet this vulnerability has largely gone unrecognized.

To address this gap, we examine the operating characteristics of thresholding rules of the form $pd >$

c under repeated use and consider how the resulting multiplicity problem can be addressed. Specifically, we propose a prior-odds correction for pd , grounded in the global null framework of Jeffreys (1938) and subsequent developments (Guo & Heitjan, 2010; Scott & Berger, 2006; Scott & Berger, 2010; Westfall et al., 1997). This approach requires specifying a prior probability for the global null hypothesis, a context-dependent quantity reflecting the a priori plausibility that all tested effects are simultaneously null. It bears emphasis that the prior-odds adjustment does not control error rates in the frequentist sense; rather, it makes the prior probability assigned to the global null explicit, so that the resulting posterior odds reflect both the data and the researcher’s substantive beliefs about the prevalence of null effects.

The paper proceeds as follows. We begin by introducing pd and examining its role as a test for existence, discussing how prior choice shapes its behavior and demonstrating why more skeptical within-model priors alone cannot resolve the multiplicity problem. We then develop the prior-odds adjustment and evaluate its properties through simulation. A worked example using the accompanying R package (Calderan & Gambarota, 2026) follows, after which we conclude with a general discussion.

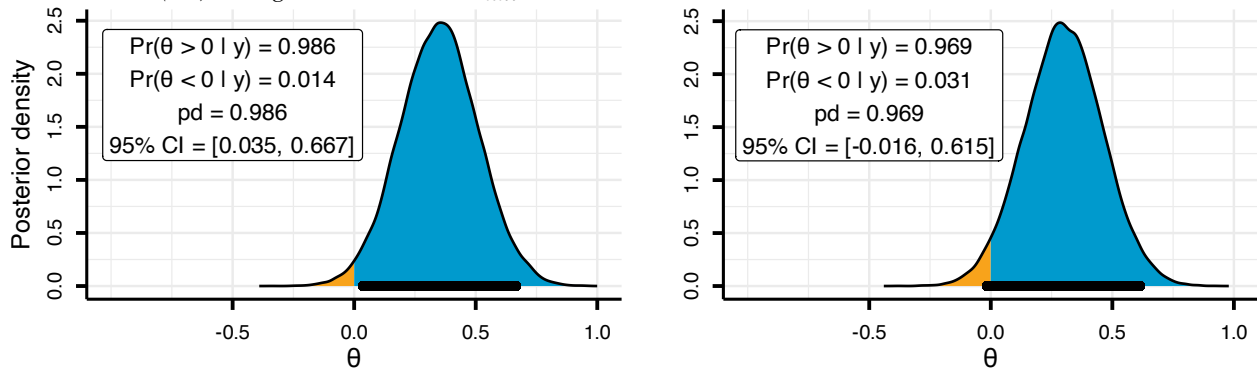
Probability of direction

The probability of direction (pd) is a posterior index that quantifies how strongly the posterior distribution favors one direction of an effect relative to a reference (θ_{null}) value (Makowski et al., 2019). For a continuous posterior, it can be defined as the larger of the two one-sided posterior probabilities: $pd = \max(\Pr(\theta < \theta_{null} | y), \Pr(\theta > \theta_{null} | y))$, and it ranges from 0.50 (no directional preference) to 1 (all on one side). Because pd is derived from the posterior, it depends on the full model specification (likelihood and prior), so the long-run error behavior of pd -based decisions is not a property of pd in isolation but of the overall Bayesian procedure that generates the posterior.

Under flat (or sufficiently diffuse symmetric) priors in typical hypothesis-testing scenarios (e.g., testing means), pd is the posterior probability of the data-favored sign relative to θ_{null} . This value corresponds to the complement of the one-sided p -value (Casella & Berger, 1987; Marsman & Wagenmakers, 2017), yielding the two-sided approximation $p \approx 2(1 - pd)$. In practice, the pd is often used to summarize both the direction and existence of an effect. This interpretation is clearest when phrased as a decision rule $pd > c$: the posterior must assign more than c probability mass to the side of θ_{null} favored by the data. Setting $c = 1 - \alpha/2$ is convenient because it bounds the posterior mass in the opposite tail by $\alpha/2$.

Figure 1

Posterior distributions illustrating the relationship between the probability of direction (pd) and 95% Credible Interval (CI), using $c = 0.975$ and $\theta_{null} = 0$.



For concreteness, suppose the posterior favors a positive effect so that $pd = \Pr(\theta > 0 | y)$. An $\alpha = 0.05$ implies $c = 0.975$, therefore the rule $pd > c$ requires that $\Pr(\theta \leq 0 | y) < 0.025$, placing less than 2.5% of posterior mass at or below 0. As Figure 1 illustrates, observing $pd > 0.975$ is equivalent to

excluding 0 from the central 95% equi-tailed credible interval. When the credible interval excludes the null value, one can infer that the null is relatively implausible (Box & Tiao, 1992). More formally, let q_p denote the p -th posterior quantile, so the central $(1 - \alpha)$ equi-tailed credible interval is $[q_{\alpha/2}, q_{1-\alpha/2}]$, which leaves probability $\alpha/2$ in each tail. The condition $\Pr(\theta \leq \theta_{\text{null}} \mid y) < \alpha/2$ holds if and only if θ_{null} falls below $q_{\alpha/2}$, that is, when $\theta_{\text{null}} \notin [q_{\alpha/2}, q_{1-\alpha/2}]$.

Given these considerations, although pd and the frequentist p -value target different quantities, it is expected that decision rules of the form $pd > c$ approximate the operating characteristics of familiar frequentist rules such as $p < \alpha$. For convenience, we express the decision threshold as $c = 1 - \alpha/2$, so that the rule $pd > c$ corresponds to excluding the null value from the central $1 - \alpha$ credible interval. In the adjusted procedure, however, c is simply a chosen posterior-probability cutoff reflecting the researcher's tolerated uncertainty before declaring an effect; the multiplicity correction is implemented entirely through the prior odds on the global null. Although we set $\theta_{\text{null}} = 0$ throughout for ease of exposition, the procedure extends readily to non-zero nulls. This is relevant, for example, when hypotheses concern a minimum effect of practical interest (Lakens et al., 2018).

Multiplicity problem

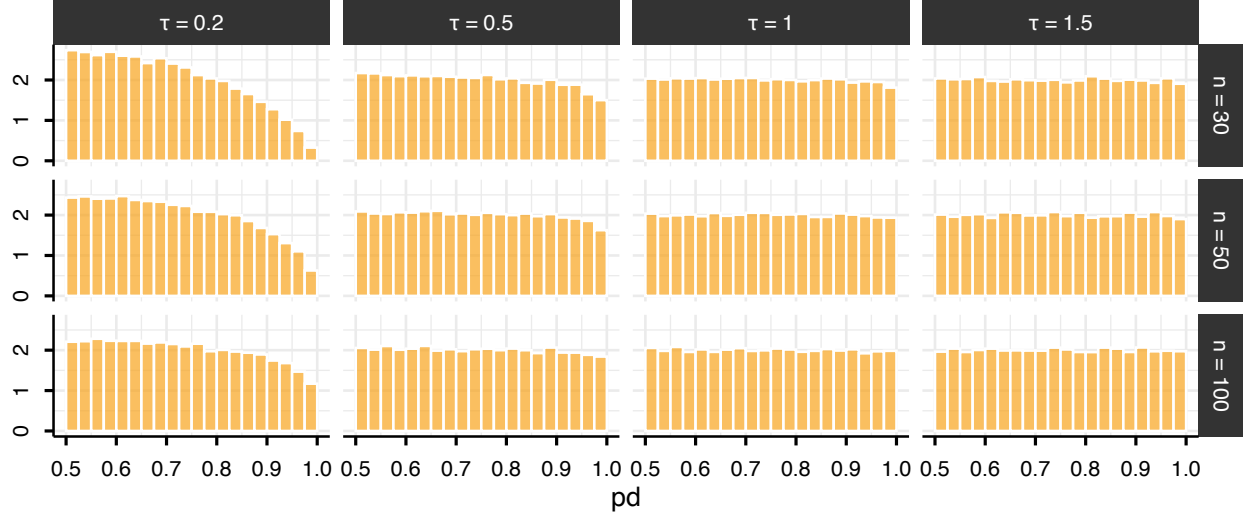
Under the null hypothesis, p -values are uniformly distributed on $[0, 1]$. Because pd can be mapped to the p -value, this implies a corresponding null reference behavior for pd on $[0.5, 1]$. However, because pd is a posterior probability regarding the sign of θ , its null distribution is not determined by the likelihood alone: it can shift depending on how strongly the prior concentrates probability near 0 (i.e., how strongly the prior “expects” near-null effects). Consequently, even in the absence of multiple testing, a threshold such $pd > 0.975$ does not correspond to a fixed rate of misleading evidence across priors and sample sizes.

To illustrate this, we simulated data under the null ($\theta = 0$) and estimated the frequency of $pd > c$, with $c = 0.975$ ($\alpha = 0.05$). Operationally, this quantifies how often the procedure falsely excludes zero from the 95% central credible interval (bounded by $1 - c$ and c), incorrectly concluding that a directed effect exists. Specifically, we conducted a Monte Carlo simulation with 10,000 replications for sample sizes $n \in \{30, 50, 100\}$. In each replication, data were drawn from a standard normal distribution, $y_i \sim \mathcal{N}(0, 1)$. For every simulated dataset, we computed the posterior distribution of θ using a conjugate normal-normal model with $y_i \mid \theta \sim \mathcal{N}(\theta, 1)$ and prior $\theta \sim \mathcal{N}(0, \tau^2)$. We varied the prior scale $\tau \in \{0.2, 0.5, 1, 1.5\}$ to encompass a range of priors, from skeptical to relatively diffuse.

As illustrated in Figure 2, under diffuse priors ($\tau \geq 1$) and/or sufficiently large sample sizes, pd exhibits a near-uniform distribution on the interval $[0.5, 1]$. Conversely, the use of a skeptical prior (e.g., $\tau = 0.2$) induces shrinkage toward 0.5, effectively reducing the error rate.

Figure 2

Empirical distributions of the probability of direction (pd) stratified by sample size ($n = 30, 50, 100$; rows) and prior scale ($\theta \sim N(0, \tau^2)$ with $\tau \in \{0.2, 0.5, 1, 1.5\}$; columns). Bin width = 0.025.



While skeptical priors reduce the probability that any single test yields an extreme pd , they do not address the issue that, as the number of tests m increases, so does the probability of observing at least one extreme value among $pd_1, \dots, pd_m$. The approach presented below addresses this at the level of the testing procedure rather than the within-model prior, by introducing a separate prior on the global null. This allows researchers using uninformative or weakly informative priors to nonetheless incorporate substantive beliefs at the family level.

As we formalize in the next section, a specific condition must be met for pd to be used in this framework. Because pd represents the posterior probability mass on one side of θ_{null} , it maps directly onto the posterior odds between directional hypotheses ($\theta > \theta_{\text{null}}$ vs. $\theta < \theta_{\text{null}}$). These posterior odds equal the Bayes Factor only when the within-model prior is symmetric around θ_{null} , ensuring that each direction is equally plausible a priori, under which the prior odds are unity and the posterior odds reduce to the ratio of marginal likelihoods. This requirement is typically satisfied by the default priors in standard Bayesian software and remains compatible with any degree of prior skepticism regarding effect magnitude.

Prior-Odds adjustment for multiplicity

From posterior probabilities to posterior odds

To connect pd to posterior odds, consider the directional hypotheses $H_+ : \theta > 0$ and $H_- : \theta < 0$. These hypotheses partition the parameter space based on the sign of θ . If the prior for θ is continuous and symmetric about 0, then the two directions are equally plausible a priori: $\Pr(H_+) = \Pr(H_-) = 0.5$.

Without loss of generality, assume the posterior favors H_+ , such that $pd = \Pr(H_+ | y) = \Pr(\theta > 0 | y)$. The complement is then given by $1 - pd = \Pr(\theta \leq 0 | y)$. The posterior odds comparing these two hypotheses are therefore:

$$\frac{pd}{1 - pd} = \frac{\Pr(H_+ | y)}{\Pr(H_- | y)}$$

For a continuous posterior distribution, the probability of any single value (e.g., $\Pr(\theta = 0 | y)$) is 0, which means that one could write $\frac{pd}{1 - pd} = \frac{\Pr(H_+ | y)}{\Pr(H_- | y)}$. However, we retain the notation H_{-0} in the denominator to explicitly indicate that the hypothesis includes the point null value of zero. Applying Bayes' rule, these

posterior odds can be decomposed into a Bayes Factor and prior odds:

$$\frac{\Pr(H_+ | y)}{\Pr(H_{-0} | y)} = \text{BF}_{+/-0} \times \frac{\Pr(H_+)}{\Pr(H_{-0})}$$

where $\text{BF}_{+/-0}$ quantifies the relative likelihood of the data under H_+ versus H_{-0} . Under prior symmetry, $\frac{\Pr(H_+)}{\Pr(H_{-0})} = 1$, which yields $\frac{pd}{1-pd} = \text{BF}_{+/-0}$ (Casella & Berger, 1987; Marsman & Wagenmakers, 2017). Thus, the odds $\frac{pd}{1-pd}$ computed from posterior draws can be interpreted directly as evidence favoring $\theta > 0$ over $\theta \leq 0$ (or $\theta < 0$ over $\theta \geq 0$).

Multiplicity adjustment via prior-odds

Assessing a single directional claim under a symmetric prior centered at θ_{null} corresponds to prior indifference: $\Pr(H_+) = \Pr(H_{-0}) = 0.5$. When many directional claims are considered jointly, however, treating each with a prior probability of 0.5 implicitly assigns near-certainty to the event that at least one effect lies in its claimed direction. A principled response is to redistribute prior probability toward the null for each individual claim as a function of m (Jeffreys, 1938; Westfall et al., 1997).

Formally, for each test $i = 1, \dots, m$, let $H_{1,i}$ denote the directional claim corresponding to the posterior-dominant sign used to compute $pd_i = \Pr(H_{1,i} | y)$, and let $H_{0,i}$ be its complement. The global null hypothesis, $\mathbf{H}_0$, is the event that none of the parameters lie in their favored directions:

$$\mathbf{H}_0 = \bigcap_{i=1}^m H_{0,i}.$$

Under exchangeability and independence, $\Pr(\mathbf{H}_0) = \pi_0^m$, so specifying $\Pr(\mathbf{H}_0)$ implies a per-test prior probability

$$\pi_0 = \Pr(\mathbf{H}_0)^{1/m}.$$

Thus, π_0 is the marginal prior probability that a given effect is null. Under the same assumptions, the number of null effects follows Binomial(m, π_0), so π_0 corresponds to the expected proportion of nulls. With π_0 in hand, the multiplicity adjustment enters entirely through the prior odds. For each test, the adjusted posterior odds satisfy:

$$\frac{pd_{\text{adj},i}}{1 - pd_{\text{adj},i}} = \frac{pd_i}{1 - pd_i} \times \frac{1 - \pi_0}{\pi_0}$$

which can be expressed as the adjusted posterior probability:

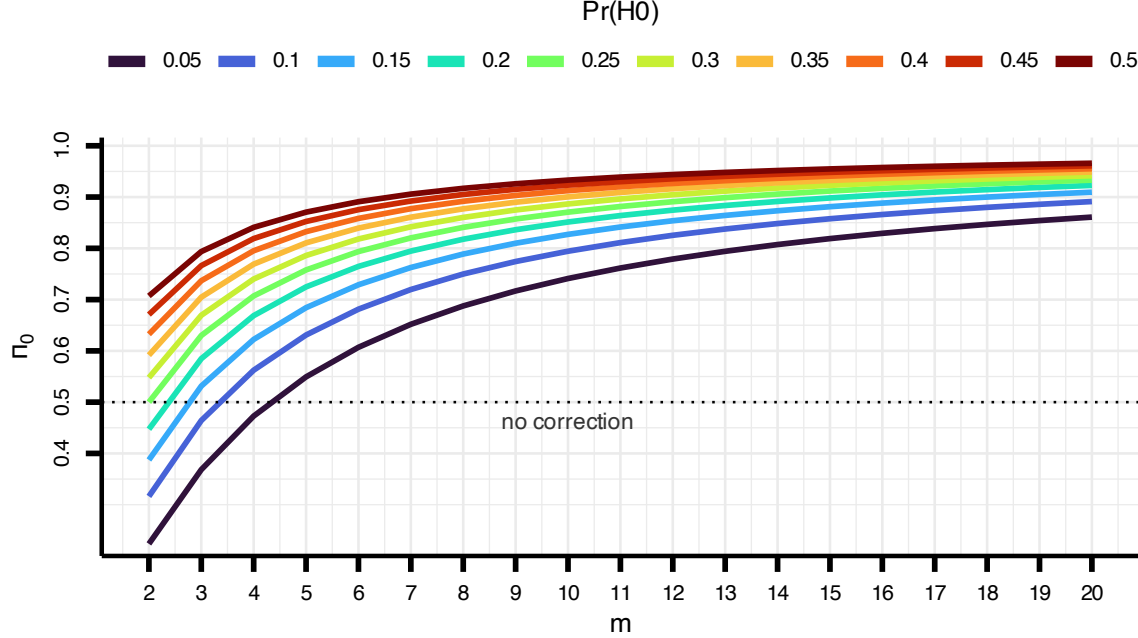
$$pd_{\text{adj},i} = \frac{pd_i \cdot (1 - \pi_0)}{pd_i \cdot (1 - \pi_0) + (1 - pd_i) \cdot \pi_0}$$

Because the prior is symmetric and pd_i is always the posterior probability of the dominant direction, the factor $(1 - \pi_0)/\pi_0$ acts as a single scalar penalty regardless of the effect's sign.

Researchers encode their expectations by choosing $\Pr(\mathbf{H}_0)$: low values are appropriate when effects are well-established, while higher values reflect greater uncertainty, as in exploratory research. For a given $\Pr(\mathbf{H}_0)$, larger families yield larger π_0 (Figure 3).

Figure 3

Impact of the multiplicity penalty. The implied prior probability $\pi_0 = \Pr(H_{0,i})$ assigned to each hypothesis $H_{0,i}$ increases as the family size m grows, depending on the specified global null probability $\Pr(\mathbf{H}_0)$. The dotted line represents the boundary condition $\pi_0 = 0.5$, below which the prior would favor H_1 a priori.



A boundary condition arises when m is small and $\Pr(\mathbf{H}_0)$ is very low, in which case the formula can yield $\pi_0 < 0.5$, inflating the posterior odds in favor of H_1 rather than penalizing them. We therefore cap π_0 at 0.5, ensuring the adjustment acts exclusively as a conservative safeguard; when no penalty is warranted, $pd_{\text{adj},i} = pd_i$.

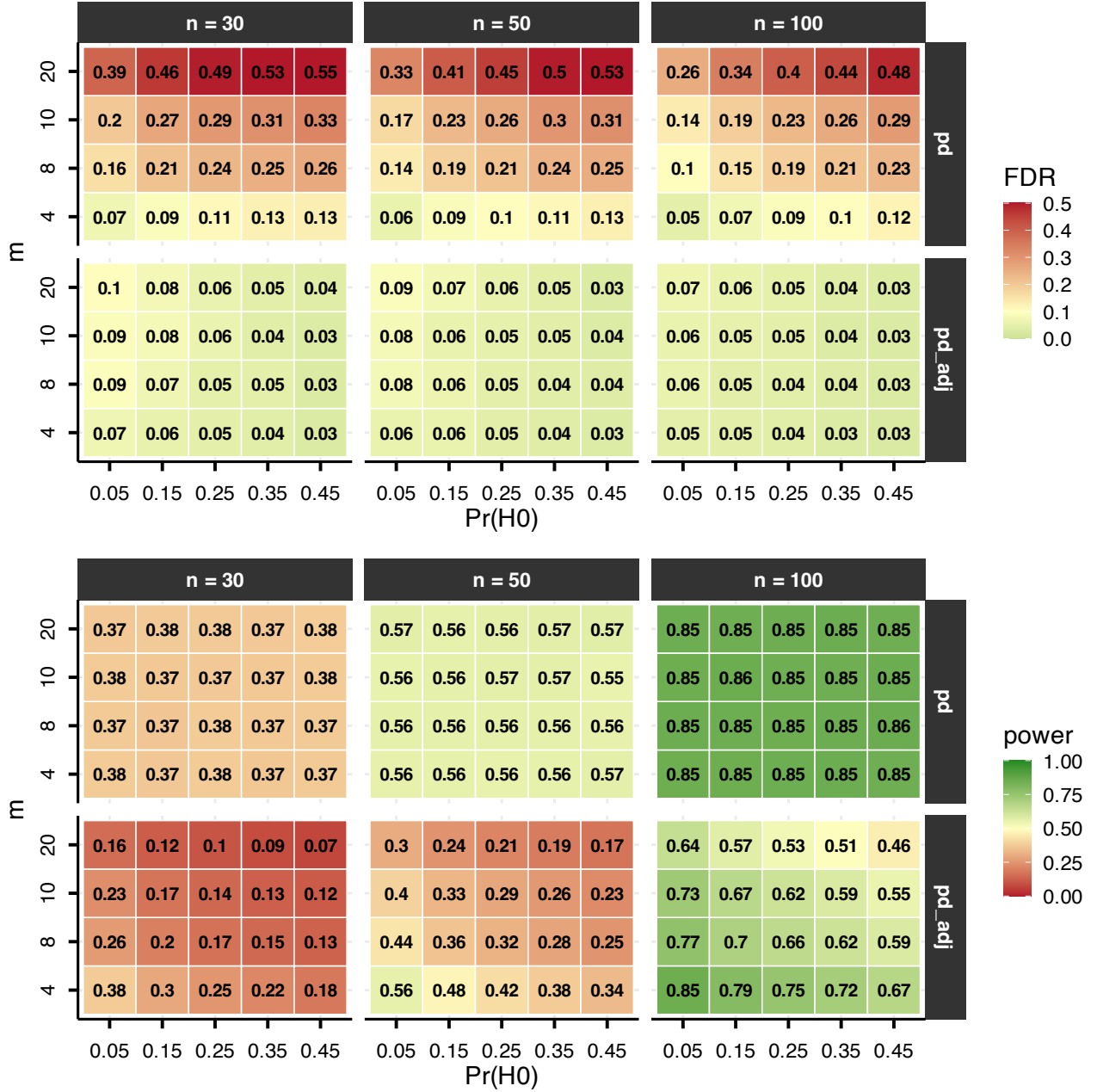
Simulation Study

To examine the impact of $\Pr(\mathbf{H}_0)$, we conducted a simulation study with a fully crossed factorial design, varying family size $m \in \{4, 6, 8, 10, 20\}$, sample size $n \in \{30, 50, 100, 500\}$, and the researcher-specified global null prior $\Pr(\mathbf{H}_0) \in \{0.05, 0.15, 0.25, 0.35, 0.45\}$, with 10,000 replications per condition. In this first study, $\Pr(\mathbf{H}_0)$ was set to match the true data-generating null proportion $\pi_{0_{\text{true}}} (\Pr(\mathbf{H}_0)^{1/m})$: at each replication, the number of true null effects was drawn from a $\text{Binomial}(m, \pi_{0_{\text{true}}})$, and the remaining hypotheses were assigned a non-null effect of $\theta_i = 0.3$. Data were drawn from a standard normal distribution, $y_i \sim \mathcal{N}(\theta_i, 1)$. For every simulated dataset, we computed the posterior distribution of θ using a conjugate normal-normal model with $y_i \mid \theta \sim \mathcal{N}(\theta, 1)$ and prior $\theta \sim \mathcal{N}(0, 2)$.

We considered the alternative hypothesis supported when $pd > 0.975$. Performance was evaluated in terms of the false discovery rate (FDR), defined as the proportion of false positives among all hypotheses declared to support the alternative, and average power, defined as the proportion of correctly identified non-null hypotheses (Benjamini & Hochberg, 1995; Izmirlian et al., 2025).

Figure 4

False Discovery Rate (FDR) and power of the unadjusted and adjusted pd procedure as a function of the prior probability of the global null ($\Pr(\mathbf{H}_0)$), the family size m , and the sample size n .

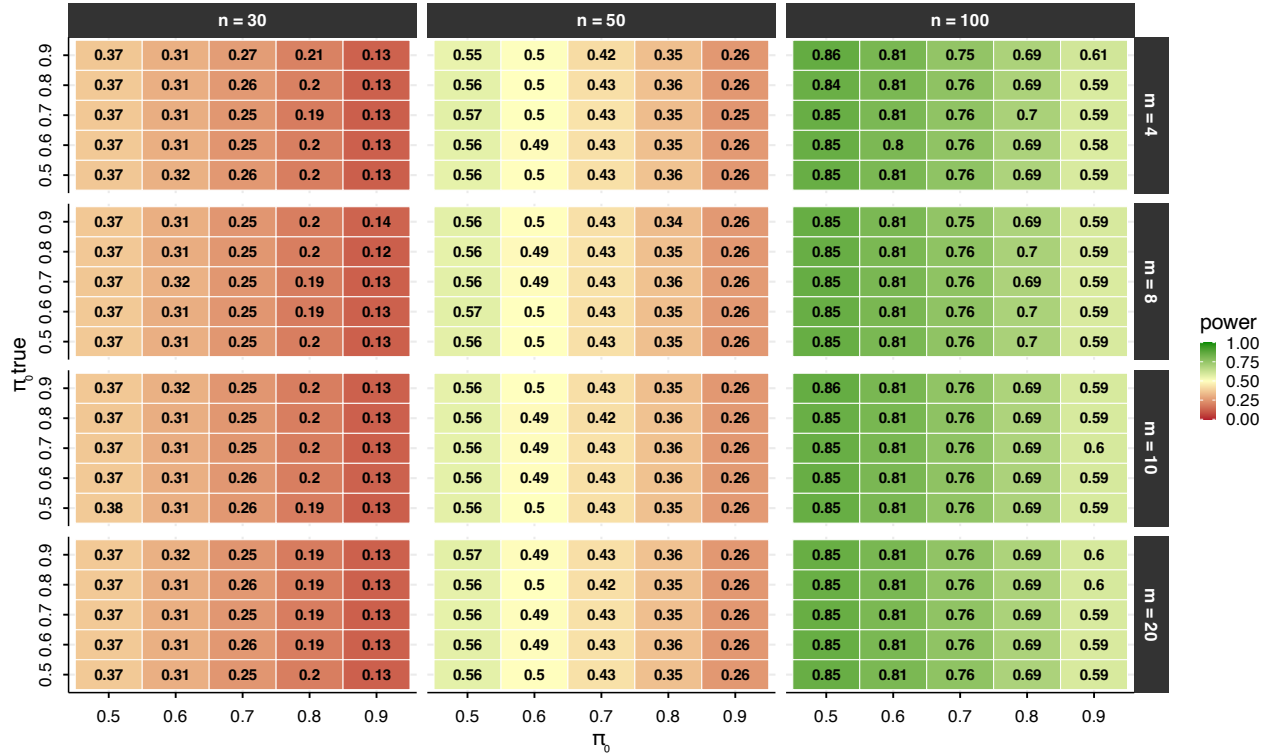


As shown in Figure 4, the unadjusted pd procedure produced substantial FDR inflation, with error rates increasing as $\Pr(\mathbf{H}_0)$ and family size m increased. By contrast, the adjusted pd procedure yielded considerably lower FDR across all conditions, with values not exceeding 0.10. Because $\Pr(\mathbf{H}_0)$ is explicitly incorporated into the correction, FDR under the adjusted procedure did not inflate as a function of the number of tests. As expected, the correction entailed a reduction in average power, with power decreasing as a function of both $\Pr(\mathbf{H}_0)$ and m reflecting the more stringent prior penalty imposed under larger families and higher global null probabilities.

Building on the previous simulation, in which the researcher-specified prior matched the true pro-

Figure 6

Average power of the adjusted pd procedure as a function of the prior probability assigned to the global null, $\Pr(\mathbf{H}_0)$, the family size m , and the sample size n . The y-axis represents π_0 , the marginal (true) probability that each individual hypothesis is null. Each cell shows the empirical power averaged over 10,000 replications.



Application: worked example with the R Package multibayes

The procedure is implemented in the `multibayes` R package, which provides functions for computing adjusted pd values from either raw posterior draws or a vector of unadjusted pd values. The core function, `pd.adjust()`, accepts posterior draws directly and computes pd values internally when needed. The output includes the adjusted pd values, the tested null value, the implied global null probability, and the number of tests m (reported as the effective number of tests when the correlation structure among parameters is accounted for). Unadjusted pd values are retained alongside adjusted ones to facilitate direct comparison; when posterior draws are provided as input, the posterior mean for each parameter is also returned.

For this example, we use the openly available dataset from Lin et al. (2020), who investigated the effect of ego depletion on conflict resolution across four studies. In their paradigm, an initial high-demand (vs. low-demand) task was followed by a Stroop task. For simplicity, we focus on Study 1, which tested the effects of congruency and ego depletion (within subjects). The dataset is readily accessible via the `afex` package (Singmann et al., 2016) and was chosen for this reason; this example is not intended as a critique of the original work. From this dataset we randomly sample 80 participants.

Given the repeated-measures structure of the data and the large number of trials per participant, we fit a multilevel generalized linear model using `brms` (Bürkner, 2017). We evaluated the effects of congruency (congruent vs. incongruent), previous congruency (congruent vs. incongruent), condition (depletion vs. control), and trial number (scaled within participant and condition) on reaction time (RT). The sequential congruency effect was modeled as the interaction between congruency and previous congruency; the hypothesis that depletion amplifies the congruency effect over time was captured by the three-way congruency

$\times$ condition $\times$ trial interaction. This yielded nine hypotheses (Table 1). A lognormal model was specified for RT. Default flat priors in brms were adopted not as a principled recommendation, but to illustrate the method's behavior under the most common applied scenario, which also represents the condition where the prior-odds adjustment is most suggested.

Although the theoretical hypotheses are directional (e.g., incongruent trials are expected to slow responses), pd is computed uniformly as $\max(\Pr(\hat{\theta} > \theta_{\text{null}}), \Pr(\hat{\theta} < \theta_{\text{null}}))$ for all hypotheses. This direction-agnostic formulation tests whether the null value falls inside or outside the posterior credible region, without presupposing a predicted direction. Note that hypotheses are tested on the log scale of the posterior. Non-zero null values were specified for the main effects of congruency, condition, and trial, as well as for the sequential congruency effect, reflecting the expectation that these effects exceed a minimal meaningful threshold rather than being merely different from zero.

Table 1

Tested hypotheses for RT, evaluated on the log scale of the posterior.

Hypothesis	Null value
H_1 : congruency effect	0.05
H_2 : previous congruency effect	0.05
H_3 : depletion effect	0.05
H_4 : trial number effect	0.05
H_5 : congruency $\times$ previous congruency interaction	0.05
H_6 : congruency $\times$ condition interaction	0
H_7 : congruency $\times$ trial interaction	0
H_8 : trial $\times$ condition interaction	0
H_9 : congruency $\times$ condition $\times$ trial interaction	0

We retained only correct trials preceded by a correct trial (i.e., error and post-error trials were excluded), yielding 5,965 observations.

```
data_rt <- data_stroop |>
  group_by(id) |>
  filter(acc == 1 & lag(acc) == 1) |>
  ungroup()

mod_rt <- brm(formula = rt ~ 1 + congruency * prev_cong +
  congruency * trial * condition +
  (1 + congruency + prev_cong +
    congruency + trial + condition || id),
  family = lognormal(), data = data_rt,
  cores = 3, chains = 3, iter = 8000)
```

Since `pd.adjust()` computes pd values automatically from a matrix of posterior draws, we extracted the relevant fixed-effect draws from the model.

```
draws_rt <- as.data.frame(mod_rt)
par_names <- colnames(draws_rt)[2:10]
draws_rt <- draws_rt[, par_names]
```

```
pd.adjust(draws = draws_rt, q = 0.02,
          null.value = c(0.05, 0.05, 0.05, 0, 0.05, 0, 0, 0, 0))
```

	mean.est	null.value	pd	pd.adj	q	m
b_congruency1	-0.1440	0.05	1.0000	1.0000	0.02	9
b_prev_cong1	-0.0166	0.05	1.0000	1.0000	0.02	9
b_trial	0.0119	0.05	1.0000	1.0000	0.02	9
b_condition1	0.0064	0.00	0.6509	0.5038	0.02	9
b_congruency1:prev_cong1	0.0222	0.05	0.9866	0.9756	0.02	9
b_congruency1:trial	-0.0033	0.00	0.7206	0.5840	0.02	9
b_congruency1:condition1	-0.0165	0.00	0.9291	0.8770	0.02	9
b_trial:condition1	0.0071	0.00	0.8936	0.8205	0.02	9
b_congruency1:trial:condition1	-0.0028	0.00	0.5961	0.5000	0.02	9
direction						
b_congruency1	two.sided					
b_prev_cong1	two.sided					
b_trial	two.sided					
b_condition1	two.sided					
b_congruency1:prev_cong1	two.sided					
b_congruency1:trial	two.sided					
b_congruency1:condition1	two.sided					
b_trial:condition1	two.sided					
b_congruency1:trial:condition1	two.sided					

The function was called with $q = 0.02$ which, with 9 test, imply an expected proportion of null ≈ 0.64 , we defined hypothesis-specific null values: $\theta_{\text{null}} = 0.05$ for congruency (H_1), condition (H_3), and cong:prev_cong (H_5), reflecting the expectation that these effects exceed a minimal meaningful threshold; $\theta_{\text{null}} = 0$ was specified for all remaining interaction terms.

```
pd.adjust(draws = draws_rt, q = 0.02,
          null.value = c(0.05, 0, 0, 0.05, 0.05, 0, 0, 0, 0),
          direction = c("greater", "greater", "less",
                        "less", "two.sided", "two.sided",
                        "two.sided", "greater", "two.sided"))
```

	mean.est	null.value	pd	pd.adj	q	m
b_congruency1	-0.1440	0.05	0.0000	0.0000	0.02	9
b_prev_cong1	-0.0166	0.00	0.0197	0.0108	0.02	9
b_trial	0.0119	0.00	0.0631	0.0354	0.02	9
b_condition1	0.0064	0.05	0.9939	0.9889	0.02	9
b_congruency1:prev_cong1	0.0222	0.05	0.9866	0.9756	0.02	9
b_congruency1:trial	-0.0033	0.00	0.7206	0.5840	0.02	9
b_congruency1:condition1	-0.0165	0.00	0.9291	0.8770	0.02	9
b_trial:condition1	0.0071	0.00	0.8936	0.8205	0.02	9
b_congruency1:trial:condition1	-0.0028	0.00	0.5961	0.5000	0.02	9
direction						
b_congruency1	greater					

b_prev_cong1	greater
b_trial	less
b_condition1	less
b_congruency1:prev_cong1	two.sided
b_congruency1:trial	two.sided
b_congruency1:condition1	two.sided
b_trial:condition1	greater
b_congruency1:trial:condition1	two.sided

Discussion

Thresholding pd into a decision rule and applying it repeatedly across many parameters induces a multiplicity problem. Without adjustment, researchers implicitly accept a prior probability for the global null hypothesis (the probability that all tested parameters are simultaneously null) that may be far lower than intended, a concern particularly acute when non-informative or weakly informative priors are used. The approach developed here addresses this by recasting multiplicity as a problem of prior specification: building on the global null formulation of Jeffreys (1938) and Westfall et al. (1997), the correction requires researchers to explicitly assign a prior probability to the global null event, separating within-model priors on parameters from prior assumptions over the hypothesis space. This makes the procedure not merely a technical correction, but a transparent mechanism for connecting prior beliefs to decision-making under multiplicity.

Related corrections have been implemented for specific testing contexts, such as post-hoc tests in JASP (Jong, 2019; van den Bergh et al., 2020). The present procedure extends this logic to pd and provides a general-purpose implementation for applied use, applicable to any posterior-derived pd or directly to posterior draws, including cases where pd values arise from multiple separate analyses within the same manuscript (Blondé et al., 2021; e.g., Ciranka et al., 2026). A key practical consideration is defining the family of hypotheses to which the adjustment applies (Hochberg & Tamhane, 1987; Shaffer, 1995). Although there is no universal consensus on what constitutes a family, we favor the criteria proposed by Westfall et al. (2011), where a family is a set of questions that “*form a natural and coherent unit, [are] considered simultaneously in the decision-making process, and/or present a set over which one ‘plays pick-the-winner.’*” This challenge is especially salient in multi-study manuscripts where the boundaries of inference are rarely made explicit, and authors should clearly specify their chosen family and explain how it serves the study’s substantive goals.

We further recommend that researchers justify their choice of prior probability for the global null by drawing on theory, expert elicitation, or domain expertise (?). The appropriate value is context-dependent: a high prior probability suits exploratory settings with many uncertain hypotheses; an intermediate value is appropriate for replication studies with moderate prior evidence; a low value is warranted in confirmatory settings with strong converging support. For applied reporting, the unadjusted pd should accompany at least one multiplicity-adjusted measure, with the number of tests and the implied global-null prior clearly stated.

Two limitations of the present framework warrant attention. First, it does not provide evidence for a point null hypothesis. The procedure evaluates directional claims defined by inequalities (e.g., $\theta > \theta_{\text{null}}$ vs. $\theta \leq \theta_{\text{null}}$), and while the complementary hypothesis includes the null value, that value receives no special status. When the null falls outside the credible interval, one can infer its relative implausibility, but this is fundamentally different from quantifying evidence in its favor (Heck et al., 2023; Ly et al., 2016; Morey & Rouder, 2011). The prior odds correction logic introduced here is nonetheless general: it applies equally to any Bayes factor method, since testing a theory requires accounting for all evidence bearing on it (Dienes, 2016). Related alternatives include combining evidence across hypotheses via joint posterior probabilities or simultaneous credible statements (Box & Tiao, 1992; Gelman & Tuerlinckx, 2000; Pruzek, 2016; Schnell

et al., 2020), as implemented for example in `credsubs` (Schnell et al., 2020); in `multibayes`, the analogous `joint()` function provides the same kind of joint-coverage output, whereas the present contribution focuses on prior-odds correction.

Second, the procedure assumes that all hypotheses are exchangeable and independent, assigning each the same prior probability of being null and ignoring dependencies among them. In practice, hypotheses are often dependent, for example when parameters are estimated jointly within the same model or when multiple tests rely on overlapping data. Bayesian approaches that explicitly model such dependence or adopt more flexible multiplicity structures have been developed (Gopalan & Berry, 1998; Guo & Heitjan, 2010; Malsiner-Walli & Wagner, 2016; Scott & Berger, 2006; Scott & Berger, 2010; Williams, n.d.), but their implementation requires substantially more modeling effort, limiting their accessibility in applied research.

As Bayesian methods continue to be adopted in psychological research, often in pragmatic settings where default priors and conventional thresholds prevail, tools that make the consequences of repeated thresholding explicit become increasingly necessary. Without multiplicity adjustment, thresholding pd across many parameters can overstate support for the alternative. The procedure developed here provides a coherent and practical way to recalibrate pd -based decisions under multiplicity, helping reported claims better reflect the strength of evidence across the full family of hypotheses.

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